HUSB238 USB-C PD Module Module

DEVLAB FACTOR

Mfr. Part #: UE0084

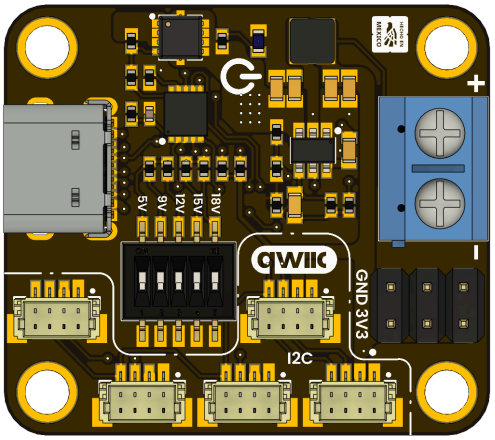
Product Reference Manual (V1.0)

Description

The DevLab HUSB238 USB-C PD Module is a USB-C Power Delivery sink and system interface designed to supply embedded platforms from modern USB-C power sources. Based on the HUSB238 Power Delivery controller, the module negotiates multiple PD voltage profiles and provides both high-voltage power outputs and a regulated 3.3 V rail for logic-level devices. Integrated I²C control and Qwiic-compatible connectors enable seamless communication and expansion, allowing the module to function as a centralized power and control node in compact and portable designs.

DevLab format compatibility.

Simplicity and compatibility are the core principles of the **DevLab form factor**. This standard defines a **compact and communication-optimized board layout**, ensuring straightforward connection and interoperability among DevLab module



Key Features

- USB-C Power Delivery sink supporting **PD 3.0**
- Wide input voltage support: **5 V, 9 V, 12 V, 15 V, 18 V, and 20 V**
- Maximum output current up to **5 A** (adapter-dependent)
- Integrated **3.3 V regulated output** for logic and sensors
- **I²C interface** for power negotiation and status monitoring
- Native **Qwiic-compatible connectors** for plug-and-play expansion
- Support for **USB Type-C v1.4**, **BC1.2**, and **Apple charging protocols**
- Compact form factor suitable for embedded and portable systems

Hardware Features

- **HUSB238 USB-C PD controller** with onboard power negotiation logic
- USB-C receptacle for power input
- High-current power output terminals for external loads
- Dedicated **3.3 V LDO/regulator** for low-power devices
- Dual **Qwiic (I²C) connectors** for sensor and module chaining
- Onboard power and status indicators
- Designed for stable operation with high-current USB-C adapters

Software Support

- I²C-based configuration and control interface
- Compatible with common microcontroller platforms (ESP32, RP2040, STM32, Arduino-compatible boards)
- Simple register-based control for voltage selection and status reading
- Suitable for use with Arduino, MicroPython, and bare-metal or RTOS-based firmware
- Example code and documentation available through the DevLab ecosystem

Applications

- USB-C power input replacement for legacy DC-powered devices
- High-power embedded systems requiring 12 V–20 V rails
- IoT nodes combining sensors and actuators with a single power source
- Robotics platforms with mixed-voltage requirements
- Portable test equipment and laboratory power modules
- Rapid prototyping and educational development platforms

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1 The Board

1.1 Accessories

2 Ratings

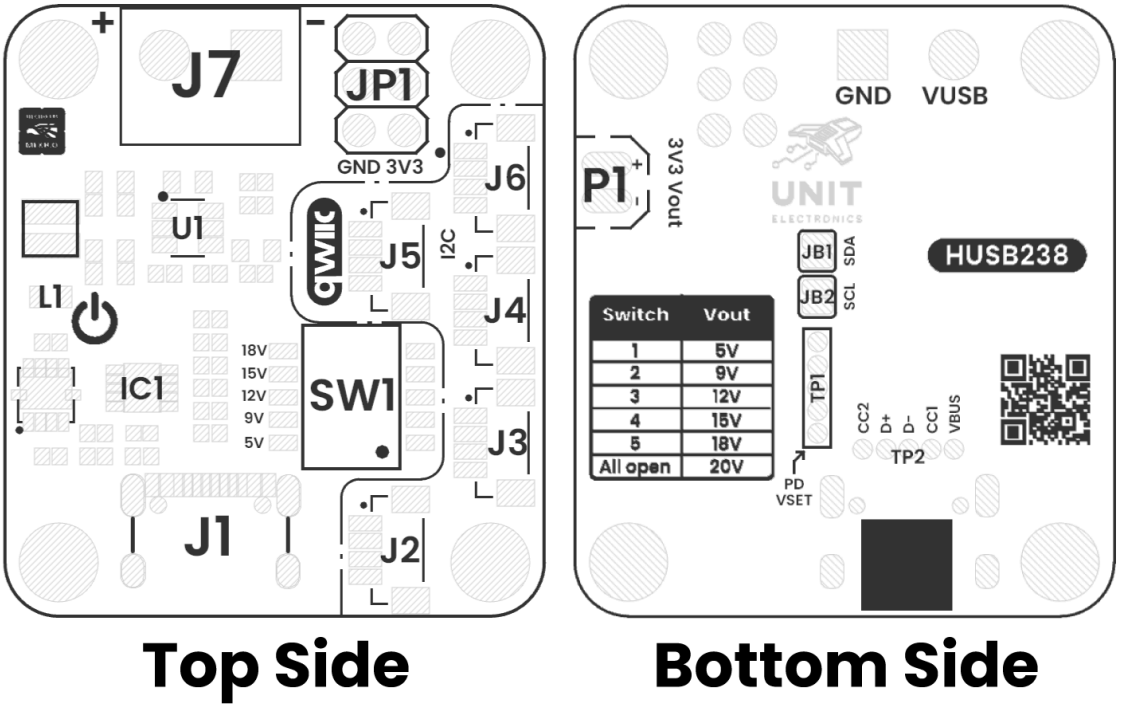
2.1 Recommended Operating Conditions

Symbol	Description	Min	Typ	Max	Unit
Vin	Input voltage to power on the module	5	-	20	V
Vih	High-level input voltage for I2C	1.4	-	5.5	V
Vil	Low-level input voltage for I2C	-	-	0.4	V
Icc	Supply Current	-	3.1	5	A
I3v3	3V3 Power Supply Output Current	-	-	2	A

3 Functional Overview

The **HUSB238 USB-C PD Module** operates as an intelligent **USB-C Power Delivery sink**, automatically negotiating power profiles from a compliant USB-C source and distributing the selected voltage to external loads while maintaining a stable **3.3 V logic rail** for control electronics.

At the core of the board, the **HUSB238 controller (IC1)** manages USB-C PD communication, advertising supported voltage profiles and selecting the requested output based on the configuration of the **DIP switch (SW1)**. This allows users to choose common PD voltages (5 V, 9 V, 12 V, 15 V, 18 V, or 20 V) without firmware, making the module suitable for both quick prototyping and production-ready designs.



The negotiated **Vout** is routed to the **VUSB screw terminal (J7)** and board pads, enabling direct connection to high-power loads such as motors, DC-DC stages, LED drivers, or embedded systems requiring higher input voltages. In parallel, an onboard **TPS54302 buck regulator (U1)** generates a regulated **3.3V supply**, exposed through the **QWIIC connectors (J2–J6)** and the **3V3/GND header (JP1)** for microcontrollers, sensors, and I²C peripherals.

For system integration and monitoring, the board provides:

- **QWIIC I²C connectivity**, with optional jumpers (**JB1/JB2**) to link the HUSB238 SDA/SCL lines directly to the QWIIC bus.
- **Test points (TP1, TP2)** for PD voltage-set diagnostics and USB signal probing.
- A **power-on LED (L1)** for immediate visual feedback of system status.
- Clearly separated **top and bottom layouts**, simplifying assembly, inspection, and debugging.

DIP Switch vs Software Control

The **HUSB238 USB-C PD Module** supports **hardware-based voltage selection** through the onboard **DIP switch (SW1)**. However, **any software or I²C command overrides the DIP switch configuration.**

Priority Order

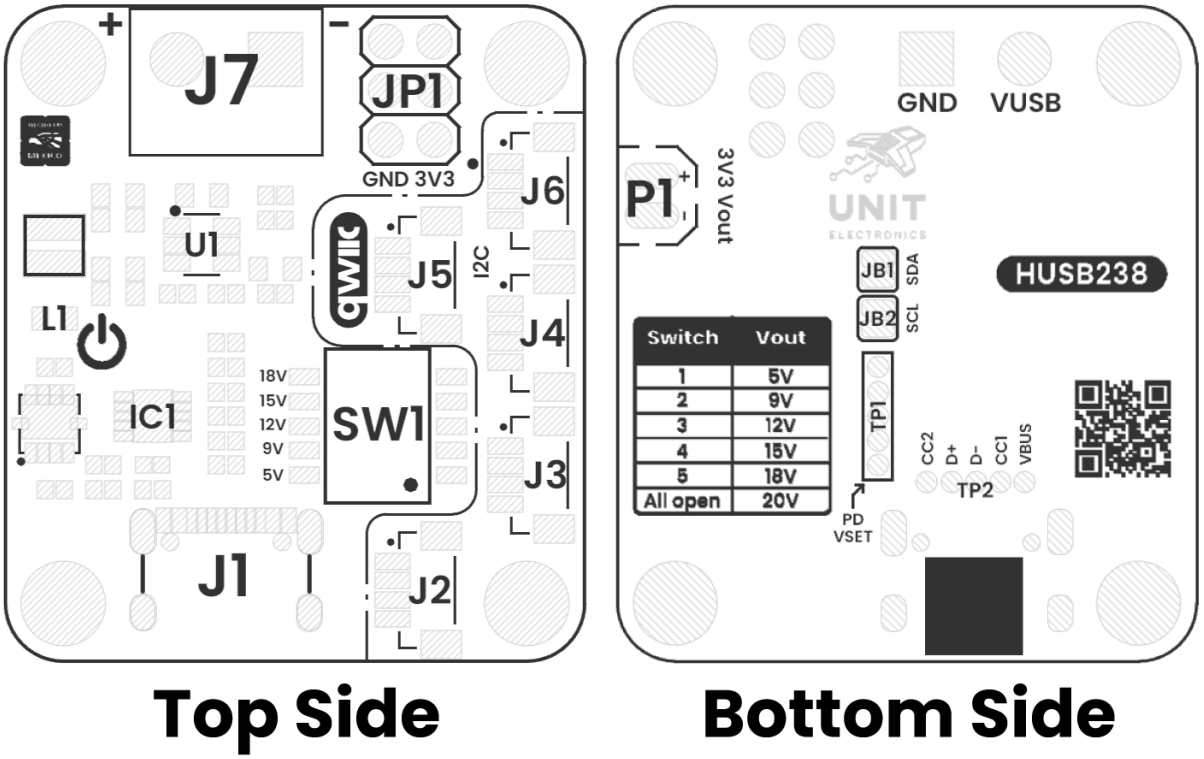
1. **Software / I²C Control (Highest Priority)**
    a. When an I²C command is issued to the HUSB238, the requested voltage profile immediately takes precedence.
    b. This allows the system to dynamically change the negotiated USB-C PD voltage during operation.
    c. Ideal for adaptive power management, smart loads, and firmware-controlled systems.

2. **DIP Switch (Startup / Fallback Priority)**
    a. The DIP switch defines the **initial voltage selection** during power-up or reset.
    b. If no software command is sent, the module operates strictly according to the DIP switch setting.
    c. Serves as a safe default and fallback configuration.

Design Rationale

- **Safety and reliability:** Hardware selection guarantees a known power state even if firmware is absent, misconfigured, or fails.
- **Ease of use:** DIP switches enable quick setup without writing code.
- **Flexibility:** Software control remains available for smart systems when hardware override is not used.

3.2 Board Topology



Top Side **Bottom Side**

Views of Board Topology

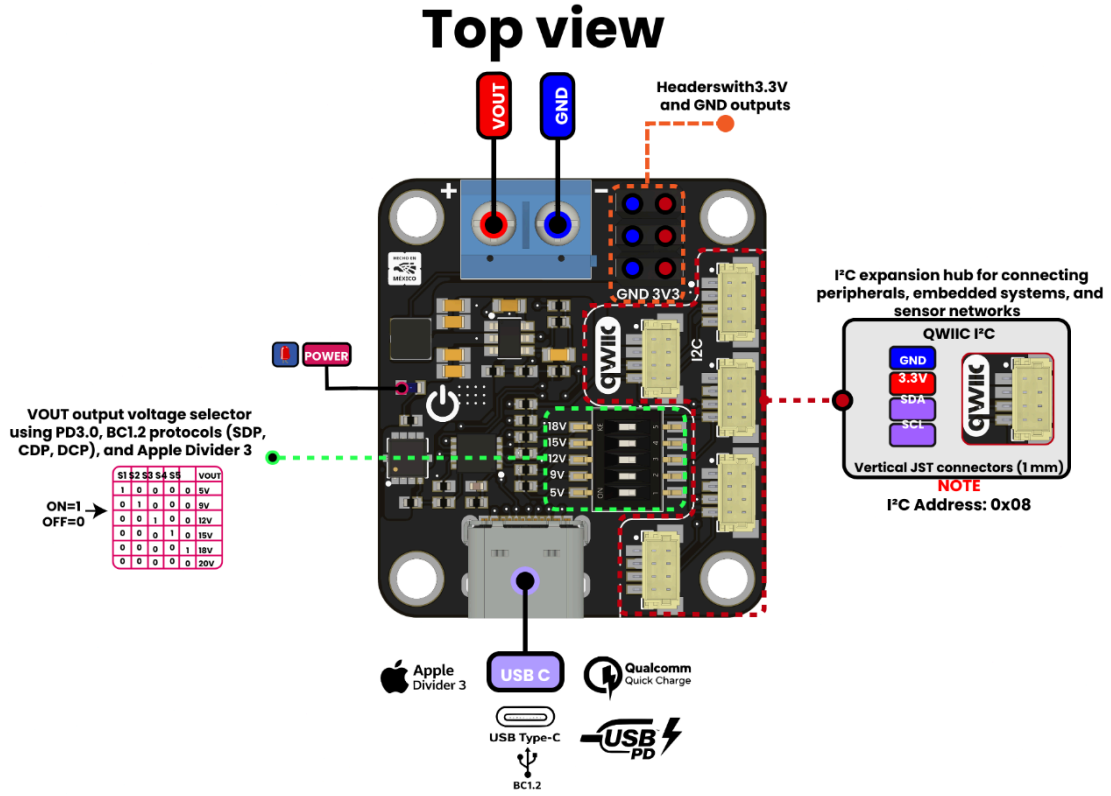
Views of DRV2605L Haptic Motor Controller Module Topology

Table 3.2.1 - Components Overview

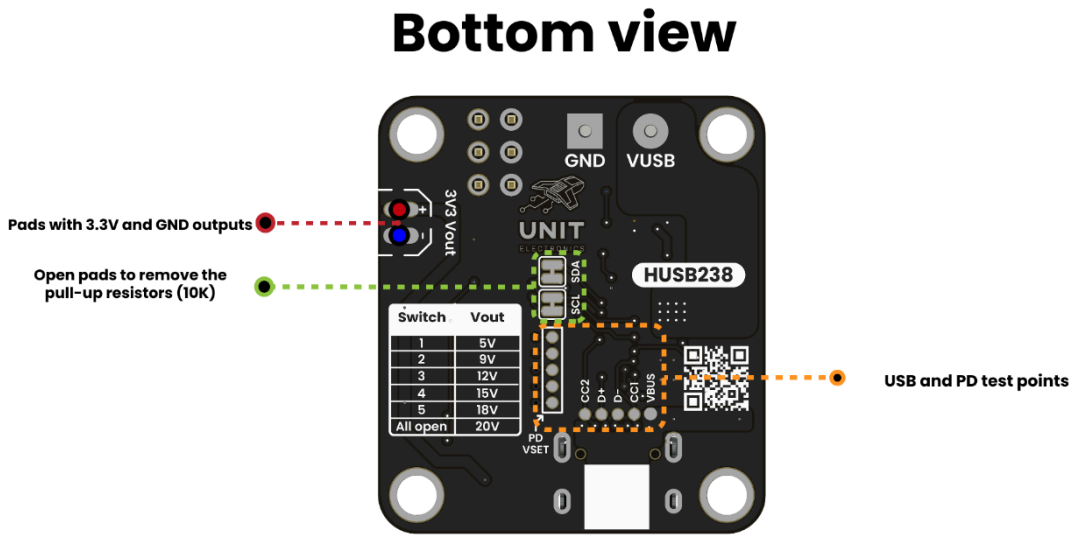
Ref.	Description
J1	USB Type-C Connector
J2-J6	QWIIC Connectors for I2C (JST 1mm)
J7	Screw Terminal Block for VUSB
L1	Power On LED
IC1	HUSB238
SW1	Dip Switch for voltage selector
U1	TPS54302 3.3V Regulator
JP1	Pin Header for 3V3 power supply and GND
JB1	Jumper Bridge to Join HUSB238 SDA with QWIIC Connectors
JB2	Jumper Bridge to Join HUSB238 SCL with QWIIC Connectors
P1	Pads for 3.3V Power Supply
TP1	Test Points for PD Vset
TP2	USB Test Points

4 Connectors & Pinouts

4.1 General Pinout



S1	S2	S3	S4	S5	VOUT
1	0	0	0	0	5V
0	1	0	0	0	9V
0	0	1	0	0	12V
0	0	0	1	0	15V
0	0	0	0	1	18V
0	0	0	0	0	20V



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DevLab HUSB238 USB-C PD Module General Pinout

4.2 Pinout General Description

Pin Label	Function	Notes
VOUT	Power Output	Configurable output voltage (5V–20V via DIP switch)
GND	Ground	Common ground reference
3V3	3.3V Output	Regulated output for external devices
SDA	I²C Data Line	Bidirectional data line (Qwiic compatible, pull-ups included)
SCL	I²C Clock Line	Clock signal for I²C communication
VUSB	USB Input Voltage	Input from USB Type-C connector
CC1 / CC2	USB Type-C Configuration	Used for USB PD negotiation
VSET	Voltage Selection Input	Controlled by DIP switch to select output voltage
QWIIC	I²C Expansion Connector	JST-SH 1.0mm connector (GND, 3V3, SDA, SCL)
3V3 Pads	Auxiliary 3.3V Output Pads	Additional access to 3.3V and GND
TEST	Test Points	USB and Power Delivery signal measurement points
SWITCH	DIP Switch	Selects output voltage (5V, 9V, 12V, 15V, 18V, 20V)
VOUT	Power Output	Configurable output voltage (5V–20V via DIP switch)
GND	Ground	Common ground reference
3V3	3.3V Output	Regulated output for external devices

5 Board Operation

5.1 Getting Started with

The DevLab: HUSB238 USB-CPD Module enables dynamic negotiation of USB Power Delivery (PD) voltages through an I²C interface. This allows embedded systems to request and monitor different voltage levels (5V–20V) from a USB Type-C power source.

This section demonstrates how to initialize the module, detect available voltages, and request a specific PD profile using an ESP32 (or compatible microcontroller).

Hardware Connections

Before powering the system, ensure the following connections:

- Connect USB Type-C to a compatible PD power source
- Connect I²C interface:
  - SDA → GPIO 6
  - SCL → GPIO 7

- Ensure GND is shared between devices

Optional: connect Serial (USB) for debugging

Required Libraries

Install the following library:

- Adafruit_HUSB238
- Wire (included in Arduino core)

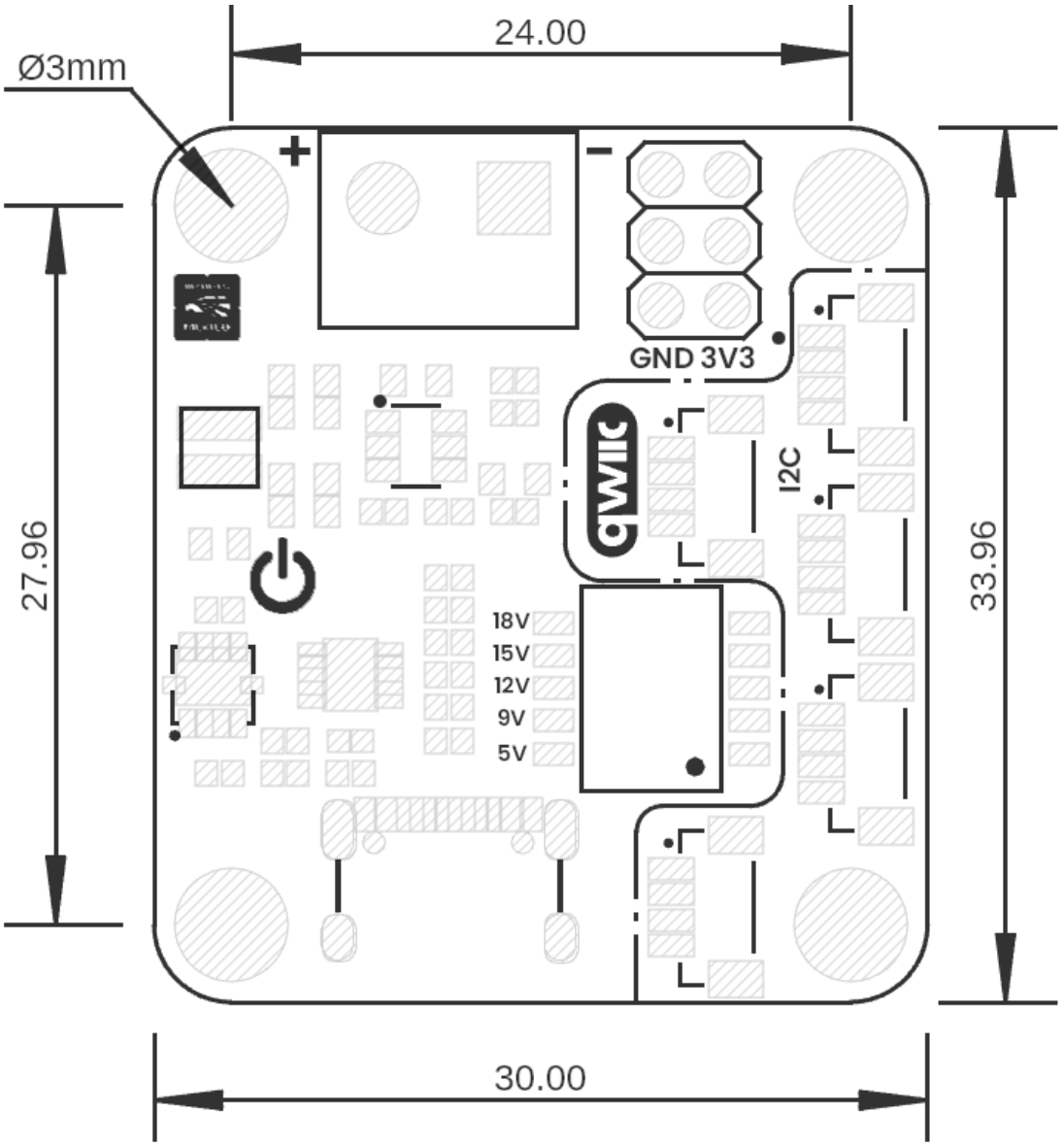
Basic Initialization Example

The following example initializes the HUSB238 module and verifies communication:

```
#include <Wire.h>
#include "Adafruit_HUSB238.h"

#define I2C_SDA 6
#define I2C_SCL 7

Adafruit_HUSB238 husb238;

void setup() {
 Serial.begin(115200);
 while (!Serial) delay(10);
```

```
  Wire.begin(I2C_SDA, I2C_SCL);

 if (!husb238.begin(HUSB238_I2CADDR_DEFAULT, &Wire)) {
   Serial.println("ERROR: No se encontró HUSB238");
   while (1) delay(100);
 }

 Serial.println("HUSB238 OK\n");
}
void loop() {
 if (!husb238.isAttached()) {
   Serial.println("Sin conexión USB-C");
   delay(2000);
   return;
 }
 // Mostrar voltajes disponibles
 Serial.println("Voltajes disponibles:");
 if (husb238.isVoltageDetected(PD_SRC_5V))  Serial.println("  5V");
 if (husb238.isVoltageDetected(PD_SRC_9V))  Serial.println("  9V");
 if (husb238.isVoltageDetected(PD_SRC_12V)) Serial.println("  12V");
 if (husb238.isVoltageDetected(PD_SRC_15V)) Serial.println("  15V");
 if (husb238.isVoltageDetected(PD_SRC_20V)) Serial.println("  20V");
  // Voltaje actual
 Serial.print("\nVoltaje actual: ");
 Serial.print(husb238.getPDSrcVoltage());
 Serial.println("V");
 // Cambiar a 15V si está disponible
 if (husb238.isVoltageDetected(PD_SRC_15V)) {
   Serial.println("\nCambiando a 15V...");
   husb238.selectPD(PD_SRC_15V);
   husb238.requestPD();
   delay(500);
   Serial.print("Nuevo voltaje: ");
   Serial.print(husb238.getPDSrcVoltage());
   Serial.println("V");
 }
  Serial.println("\n---\n");
 delay(5000);
}
```

6 Mechanical Information



Board Dimensions in Millimeters

Mechanical dimensions in millimeters

7 Company Information

Company name	UNIT Electronics
Company website	https://uelectronics.com/
Company Address	Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX

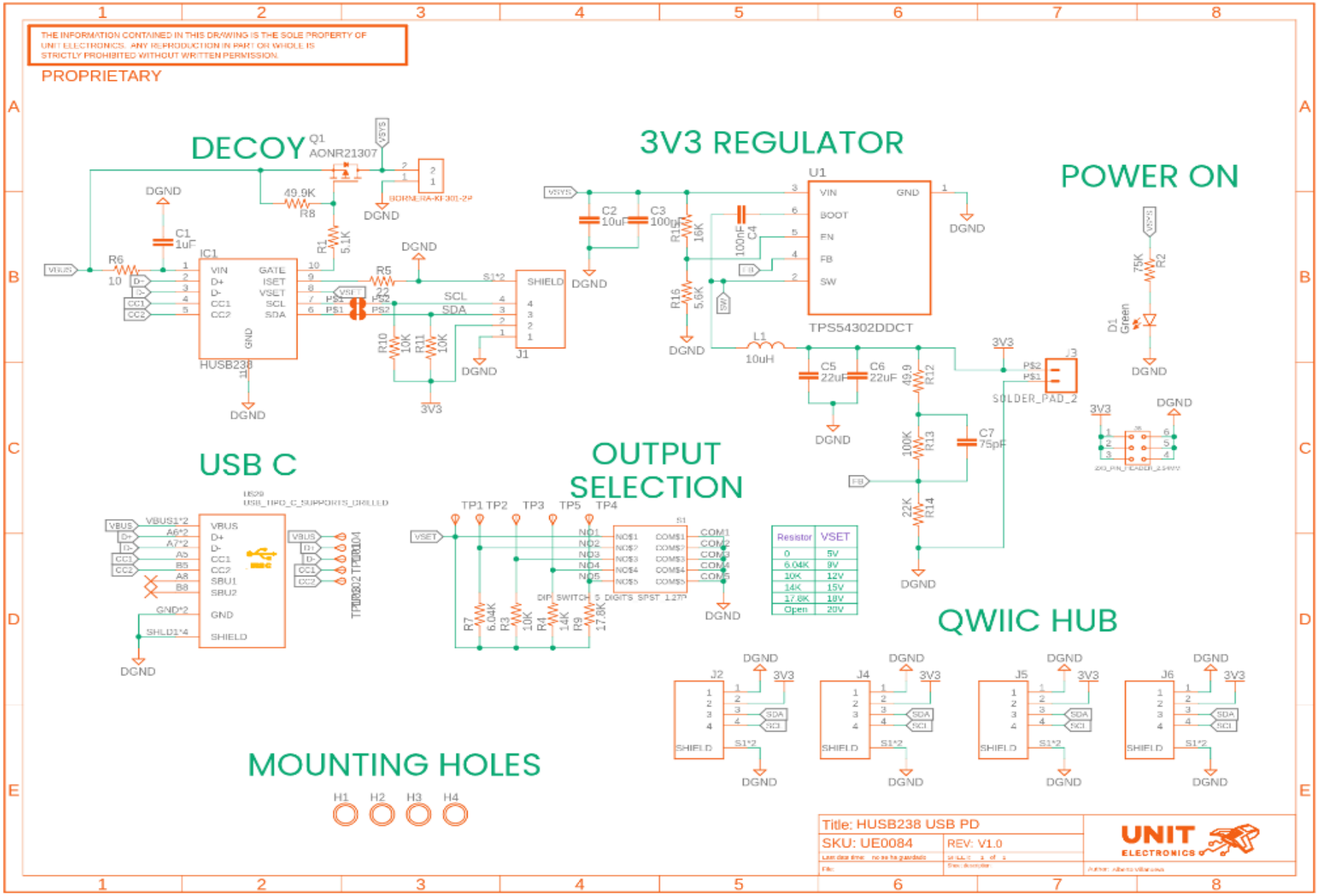
8 Reference Documentation

Ref	**Link**
Documentation	https://github.com/UNIT-Electronics-MX/unit_devlab_husb238_usb_c_pd_module
Getting Started Guide	https://wiki.uelectronics.com/wiki/devlab-husb238-usb-c-pd-module
Thonny IDE	https://thonny.org/
Arduino IDE	https://www.arduino.cc/en/software
Visual Studio Code	https://code.visualstudio.com/download

9 Appendix

9.1 Schematic

(*https://github.com/UNIT-Electronics-MX/unit_devlab_husb238_usb_c_pd_module/blob/main/hardware/unit_sch_v_1_0_0_ue0084_devlab_husb238_usb-c_pd_module.pdf*)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PROPRIETARY
DECOY
3V3 REGULATOR
POWER ON
USB C
OUTPUT SELECTION
QWIIC HUB
MOUNTING HOLES
Q1 AONR21307
IC1
HUSB238
U1
TPS54302DDCT
J1
J2
J3
J4
J5
J6
SOLDER_PAD_2
DIP SWITCH 5 DIGITS SPST 1.27P
Resistor | VSET
0 | 5V
6.04K | 9V
10K | 12V
14K | 15V
17.8K | 18V
Open | 20V
H1 H2 H3 H4
Title: HUSB238 USB PD
SKU: UE0084
REV: V1.0
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